

Study Design, Multimodal Data Harmonization, and Explainable ML Architecture

From Multi-Center Registry Cohort to Nested Bayesian Validation and Bedside XAI Explanations

Step 1: Patient Cohort

(N = 350 Patients)

Inclusion Criteria:

- Genetically confirmed 5q SMA (homozygous SMN1 deletion)
- Clinical Type 1 (n = 154) or Clinical Type 2 (n = 196)
- Documented SMN2 copy number via MLPA (2, 3, or 4 copies)
- Baseline CMAP electrophysiology
- Complete 24-month longitudinal motor function follow-up

Treatment Regimens:

- Nusinersen: 158 (45.1%)
- Risdiplam: 117 (33.4%)
- Onasemnogene: 57 (16.3%)
- Untreated Controls: 18 (5.1%)

Assessment Intervals:

Baseline (0m), 6m, 12m, 18m, and 24m follow-up.

Target: 24-Month Delta Score and MCID Attainment.

Step 2: Feature Space

(11 Multimodal Variables)

Genomic Predictor:

- SMN2 Copy Number (2, 3, 4)

Chronological Predictors:

- Age at Symptom Onset (mo)
- Treatment Initiation Delay (mo)
- Chronological Age at Tx (mo)

Neurophysiology:

- Baseline Ulnar CMAP (mV)

Motor Function Scale:

- Raw Baseline Score
- Normalized Baseline Motor (%) [CHOP-INTEND or HFMSE]

Anthropometric & Clinical:

- Nutritional Z-Score (BMI/WFA)
- Daily Baseline NIV (hrs/day)
- Scoliosis Cobb Angle (deg)

Preprocessing Pipeline:

Median/MICE imputation (<2.5%)
Robust scaling & one-hot encoding.

Step 3: ML Modeling

(Nested Cross-Validation)

- Gradient Boosting (GBM) [Best]
- Random Forest (RF)
- Candidate Architectures (SVR)
- Multi-Layer Perceptron (MLP)
- ElasticNet Regression
- Ridge Regression

Validation Rigor:

- 10-Fold Outer Loop (Eval)
- 5-Fold Inner Loop (Bayesian hyperparameter tuning)

- Stratification across SMA Type 1 and Type 2 cohorts

- Permutation testing (n = 1000) and Monte Carlo stability splits

Primary Benchmark:

GBM achieves R2 = 0.840, RMSE = 1.77, Pearson r = 0.926.

Step 4: Explainable AI

(Game-Theoretic SHAP)

Global Feature Attributions:

- Exact TreeExplainer Shapley values for all predictions
- Global ranking: Treatment Delay (#1) and CMAP (#2) dominate
- Beeswarm directional plots

Feature Interactions:

- Identified synergies with second-order interaction effects
- Critical therapeutic window: Delay <= 3.5 mo drives +2.8 pts

Bedside Decision Support:

- Partial dependence waterfall decompositions
- Transparent counterfactuals
- FHIR / EHR integrable

Clinical Impact:

Objective prognostic counseling and optimized triage for early DMT.